

Mengqi LI

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Ph.D. Student in Geography

RESEARCH INTERESTS

- GeoAI
- Human Mobility
- Disaster Resilience

EDUCATION

- **University at Buffalo, SUNY** Aug. 2025 - Present
Ph.D. in Geography (Specialization in GeoAI & Disaster Resilience)
Buffalo, NY
 - **Advisor:** Prof. Dr. Yingjie Hu
- **University of Zurich** Sep. 2023 - Jun. 2025
M.S. in Geographic Information Science and Systems
Zurich, Switzerland
 - **Advisor:** Prof. Dr. Cheng Fu, Prof. Dr. Robert Weibel
 - GPA: Summa cum laude
- **Nanjing University of Information Science & Technology** Sep. 2019 - Jun. 2023
B.S. in Geographic Information Science
Nanjing, China
 - GPA: 4.025/5 (Outstanding graduate)

PEER-REVIEWED PUBLICATION

- [1] **Li, M.**, Dai, W., Wang, G., Wang, B., Gao, Y., Amankwah, S. O. Y., "Reconstructing High-Resolution DEMs from 3D Terrain Features using Conditional Generative Adversarial Networks". In: *International Journal of Applied Earth Observation and Geoinformation* (2024). DOI: [10.1016/j.jag.2024.104115](https://doi.org/10.1016/j.jag.2024.104115)
- [2] Luo, W., Liu, Z., Ran, Y., **Li, M.**, Zhou, Y., Hou, W., Lai, S., "Unraveling varying spatiotemporal patterns of Dengue Fever and associated exposure-response relationships with environmental variables in three Southeast Asian countries before and during COVID-19". In: *PLOS Neglected Tropical Diseases* 19.4 (2025), e0012096
- [3] Qiu, R., Dai, W., Wang, G., Luo, Z., **Li, M.**, "Evaluation of different deep learning methods for meteorological element forecasting". In: *IEEE Access* (2024). DOI: [10.1109/ACCESS.2024.3411109](https://doi.org/10.1109/ACCESS.2024.3411109)
- [4] **Li, M.**, Dai, W., Fan, M., Qian, W., Yang, X., Tao, Y., Zhao, C., "Combining Deep Learning and Hydrological Analysis for Identifying Check Dam Systems from Remote Sensing Images and DEMs in the Yellow River Basin". In: *International Journal of Environmental Research and Public Health* 20.5 (2023), p. 4636. DOI: [10.3390/ijerph20054636](https://doi.org/10.3390/ijerph20054636)
- [5] Chen, K., Wang, C., Lu, M., Dai, W., Fan, J., **Li, M.**, Lei, S., "Integrating Topographic Skeleton into Deep Learning for Terrain Reconstruction from GDEM and Google Earth Image". In: *Remote Sensing* 15.18 (2023), p. 4490. DOI: [10.3390/rs15184490](https://doi.org/10.3390/rs15184490)

SKILLS

- **Programming & ML:** Python, SQL, R, PyTorch, TensorFlow, Scikit-learn
- **Data Science Tools:** Pandas, NumPy, Jupyter, Git, Linux, PostgreSQL
- **Geospatial Tech:** Google Earth Engine (GEE), ArcGIS Pro, QGIS

RESEARCH & PROJECT EXPERIENCE

- **GeoAI for Wildfire Smoke Exposure Modeling** Aug. 2025 - Present
Graduate Research Assistant, University at Buffalo
Buffalo, NY
 - Generated high-resolution hourly wildfire smoke by leveraging low-cost sensor networks and ML models.
 - Quantified dynamic smoke exposure by integrating human mobility data, revealing how human movements alter.
 - Evaluated inequalities by linking exposure disparities with socioeconomic vulnerability across different demographic groups.
- **GeoAI-Driven Mapping of Informal Settlements via Multi-Modal Prompting** Nov. 2025 - Present
Joint Research with Texas A&M, HKUST(GZ), ETH Zürich, & CAU
Remote / Project leader
 - Engineered a ML framework to extract individual building footprints within highly dense urban clusters.
 - Integrated multi-source social sensing data to capture human activity patterns to guide the identification of slums.

- Bridged physical urban morphology with socioeconomic dynamics.
- **Satellite Imagery Analysis: 3D Building Evolution** Oct. 2023 - Sep. 2024
Research Assistant, ETH Zürich Zurich, Switzerland
 - Applied deep learning models on satellite imagery to track 3D building-stock evolution in urban China.
 - Built automated data pipelines using Planet-API and Google Earth Engine for acquiring and pre-processing large-scale remote sensing imagery.
 - Created high-quality training datasets for semantic segmentation models.
- **Computer Vision: 3D Terrain Reconstruction with GANs** Oct. 2022 - May 2023
Bachelor Thesis Project Nanjing, China
 - Developed a Conditional Generative Adversarial Network (CGAN) framework to reconstruct high-resolution 3D terrain models from sparse features.
 - Achieved a 71.96% improvement in reconstruction accuracy compared to traditional interpolation methods.
 - Designed custom loss functions to optimize edge detection and topological consistency in generated images.

PROFESSIONAL EXPERIENCE

- **University at Buffalo** Jan. 2026 - Present
Teaching Assistant Buffalo, NY
 - GEO 481/506 Geographical Information Systems Lab
- **ETH Zürich** Oct. 2023 - Sep. 2024
Research Assistant Zurich, Switzerland
 - Applied deep learning models on satellite imagery to track 3D building-stock evolution in urban China.
 - Built automated data pipelines using Planet-API and Google Earth Engine for acquiring and pre-processing large-scale remote sensing imagery.
 - Created high-quality training datasets for semantic segmentation models.
- **Esri (China) Information Technology Co., Ltd.** Jun. 2021 - Sep. 2022
Trainee Technical Engineer Beijing, China
 - Analyzed enterprise bid data using web crawling techniques.
 - Independently drafted a successful tender for ArcGIS Desktop software.
 - Tested new ArcGIS Pro functions and authored User Guidance documentation.

SELECTED AWARDS

- Outstanding Graduates Jun. 2023
- First Class Scholarship 2020-2023
- Silver Award in Esri-Cup College Students GIS Software Development Contest Oct. 2021

REFERENCES

Available upon request.